ProveIt / Analysis / Polylogarithms

Algebraic Gauss Residues and Centered Harmonic Reflection

Finite Stirling formulas, all-depth trivial zeros, Gamma generating functions, and Stieltjes identities

Research continuation prepared for the ProveIt project

10 October 2026

Abstract

An explicit research question in the repository asks for a finite algebraic proof of a Bernoulli-coefficient identity previously obtained from residues of Gauss summation. We give such a proof, using integer-root interpolation and a degree argument. We then evaluate the entire resonant coefficient tail: its normalized exponential generating function is $e^{ct}{}_2F_1(a, b; r + 1; 1 - e^t)$, and every coefficient has a finite Stirling-number formula. The coefficients form an Appell family in the comparison shift.

A second development uses the moving reflection $a \mapsto 1 + u - a$ in the complex-order Barnes–Hurwitz family. Its centered harmonic derivatives have a common holomorphic zero at every nonpositive even integer. We prove that the centering is uniquely forced by the corresponding all-depth condition; for affine shifts, only the first two depths are needed. An exact Gamma quotient evaluates an infinite family of convergent harmonic sums at outer exponent two, with an explicit zeta-cumulant formula at every depth. We also give shifted, all-order logarithmic polygamma sums involving generalized Stieltjes constants and Hurwitz zeta derivatives, ordinary antiderivatives, and an unequal-frequency multivariate reflection theorem.

The existing fixed-shift Laurent theorem is acknowledged as prior work, not relabeled as a new result. A version-specific Stieltjes indexing error in a cited preprint is corrected. All-index claims are proved analytically or algebraically. The accompanying 436 exact checks and 30 numerical comparisons are separate regression evidence. Global priority is not asserted, and neither S6 nor S8 is proved here.

Inspected scientific baseline: `dd78c6c9079e1e4a32600d8f9400667b4de8bd54`.
The report and code are self-contained; a proposed integration map and source-audit notes accompany the article.

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1 Scope, provenance, and principal conclusions

The repository's polylogarithm programme already includes a substantial calculus of Hurwitz jets, harmonic Dirichlet series, Gamma quotients, Stieltjes constants, and regularized correlations. Its current reading artifact distinguishes accepted scientific material from queued incoming reports [1]. That distinction is retained here.

The most specific problem addressed is the request in the *Stieltjes–harmonic identities* research agenda for a direct finite Bernoulli-polynomial proof of the Gauss residue identity [2, 3]. The identity was already proved there analytically; the open task was an independent algebraic explanation. Theorem 3.2 supplies that explanation. More importantly, Theorem 4.1 evaluates every coefficient after the residue, not only the residue itself. This is the main algebraic extension.

The harmonic part starts from a different preserved report, whose all-depth, outer-shifted Laurent theorem already determines the principal and constant terms at nonpositive integers [4]. We use that theorem as established background and give a short independent local argument where needed. The extension developed here is the *moving* reflection and centering: the shift varies with the harmonic-order parameter before coefficients are extracted. This produces common holomorphic zeros of all depths, an elementary uniqueness theorem, a Gamma generating function for convergent sums, and an unequal-frequency multivariate version.

The generating Barnes–Hurwitz family itself is classical. Young's height-one generating theorem [5] is a direct antecedent; related Laurent and harmonic-zeta work includes [6, 7]. Standard Bernoulli and Hurwitz conventions are those in [8, 9]. We do not infer global originality from failure to locate an identical formula. The new statements are presented as proved continuations relative to the inspected textual sources, with an explicit overlap map in the accompanying audit.

1.1 What is proved, and what is not

The article establishes the following interconnected conclusions.

1. An elementary residue proof and a finite formula for the full resonant Gauss tail, including shift transport and mixed parameter jets.
2. A moving-reflection theorem, common all-depth trivial zeros, and uniqueness of the centered affine shift.
3. An every-depth zeta-cumulant evaluation of convergent harmonic sums; a compatible centered Stieltjes normalization; and all-order shifted polygamma–logarithm identities with ordinary antiderivatives.
4. A reflection theorem for any finite number of positive, unequal frequencies and independent harmonic-depth parameters.

No claim is made that individual colored polylogarithms reduce to depth one, that all higher harmonic Stieltjes coefficients have finite Gamma reductions, or that S6 or S8 vanishes. The incoming directory was inspected as a queue; its binary archives were not all unpacked and audited in this run. This is not a completeness claim about that queue.

2 Conventions and the harmonic Barnes family

For $n, r \geq 0$, put

$$E_{n,r} = [u^r] \prod_{j=1}^{n} \left(1 + \frac{u}{j}\right), \qquad E_{n,0} = 1. \tag{2.1}$$

An empty product is one. Thus

$$E_{n,1} = H_n,$$
$$E_{n,2} = \frac{H_n^2 - H_n^{(2)}}{2},$$
$$E_{n,3} = \frac{H_n^3 - 3H_nH_n^{(2)} + 2H_n^{(3)}}{6}, \qquad H_n^{(j)} = \sum_{k=1}^{n} k^{-j}. \tag{2.2}$$

With $\Re a > 0$, define

$$\mathcal{B}(s,u;a) = \sum_{n=0}^{\infty} \frac{(1+u)_n}{n!(n+a)^s}, \qquad Z_r(s,a) = [u^r]\mathcal{B}(s,u;a). \tag{2.3}$$

The series for Z_r converges normally for $\Re s > 1$; the bivariate series converges normally on compact subsets of, for example, $|u| < \delta$, $\Re s > 1+\delta$, $\Re a > 0$. All powers of $n+a$ use the principal logarithm. At $a = 1$, $Z_r(s,1) = \zeta(s,\{1\}^r)$ in the convention with the largest index first.

Lemma 2.1 (Mellin representation and uniform subtraction)**.** *In the common convergence domain,*

$$\Gamma(s)\mathcal{B}(s,u;a) = \int_0^{\infty} t^{s-1}\mathrm{e}^{-at}(1-\mathrm{e}^{-t})^{-1-u}\,\mathrm{d}t. \tag{2.4}$$

Define the analytic heat coefficients at $t = 0$ by

$$\phi(t,u,a) = \mathrm{e}^{-at}\left(\frac{t}{1-\mathrm{e}^{-t}}\right)^{1+u} = \sum_{j\geq 0} c_j(u,a)t^j. \tag{2.5}$$

For every $K \geq 1$,

$$\Gamma(s)\mathcal{B}(s,u;a) = \sum_{j=0}^{K-1} \frac{c_j(u,a)}{s-1-u+j} + J_K(s,u;a), \tag{2.6}$$

where J_K is jointly holomorphic for $\Re(s-u) > 1-K$, locally in $\Re a > 0$. Increasing K continues the same meromorphic family.

Proof. The binomial series is $\sum_{n\geq 0}(1+u)_n\mathrm{e}^{-nt}/n! = (1-\mathrm{e}^{-t})^{-1-u}$. Termwise integration is justified by absolute convergence and the Laplace integral for $(n+a)^{-s}$. Split the integral at $t = 1$. On $(0,1)$, subtract the first K terms of (2.5); the remainder is $O(t^K)$ uniformly on compact parameter sets. The remaining small-t integrand is bounded by a constant times $t^{\Re(s-u)-2+K}$. Parameter derivatives add only fixed powers of $\log t$ and preserve integrability on smaller compact subsets. At infinity the positive real part of a gives exponential decay, also after differentiation. Integrating the subtracted monomials gives (2.6). The formulas agree in the original half-plane, so their continuations are compatible. □

The heat coefficients are rational polynomials in u, a. A convenient exact recursion follows from

$$\log\phi(t,u,a) = \left(\frac{1+u}{2} - a\right)t - (1+u)\sum_{k\geq 1}\frac{B_{2k}}{2k(2k)!}t^{2k}. \tag{2.7}$$

This is an analytic germ, not an asymptotic series. The Bernoulli numbers use $B_1 = -1/2$.

We will also use the standard Laurent convention

$$\zeta(1+\varepsilon,a) = \varepsilon^{-1} + \sum_{j\geq 0}\frac{(-1)^j\gamma_j(a)}{j!}\varepsilon^j, \qquad \gamma_0(a) = -\psi(a). \tag{2.8}$$

A finite part always denotes a Laurent coefficient of a specified meromorphic function; it is not an ordinary sum of a divergent series.

3 A finite algebraic proof of the Gauss residue identity

The coefficients in this section are different from the heat coefficients c_j above. For indeterminates a, b, s, c, define

$$D_j(a,b,s;c) = \frac{(-1)^{j+1}}{j(j+1)}\big(B_{j+1}(a-c) + B_{j+1}(b-c) - B_{j+1}(a+b+s-c) - B_{j+1}(1-c)\big), \tag{3.1}$$

$$\sum_{m\geq 0} A_m(a,b,s;c)t^m = \exp\left(\sum_{j\geq 1} D_j(a,b,s;c)t^j\right). \tag{3.2}$$

The last identity is formal. In particular,

$$A_0 = 1, \qquad mA_m = \sum_{j=1}^{m} jD_jA_{m-j}. \tag{3.3}$$

For every fixed m, this is a finite polynomial definition over $\mathbb{Q}$.

Lemma 3.1 (Exact comparison-shift transport)**.** *For $m \geq 1$,*

$$\partial_c A_m(a,b,s;c) = (s+m)A_{m-1}(a,b,s;c). \tag{3.4}$$

Consequently, for an arbitrary shift h,

$$A_m(a,b,s;c+h) = \sum_{j=0}^{m}\binom{s+m}{j}h^jA_{m-j}(a,b,s;c). \tag{3.5}$$

The binomial coefficient here is its polynomial meaning.

Proof. The Bernoulli derivative formula gives $\partial_c D_1 = s+1$ and $\partial_c D_j = (j-1)D_{j-1}$ for $j \geq 2$. Writing the formal series in (3.2) as $G(t)$ gives $\partial_c G = (s+1)tG + t^2\partial_t G$. Coefficient comparison proves (3.4). Iterating that derivative and applying the finite Taylor formula proves (3.5). □

At $s = -r$, the coefficient A_r is already seen to be independent of c. The following argument determines it without Gamma functions, infinite sums, or meromorphic residues.

Theorem 3.2 (Algebraic Gauss residue identity)**.** *For every integer $r \geq 0$ and all complex a, b, c,*

$$\boxed{A_r(a,b,-r;c) = \frac{(-1)^r}{r!}(1-a)_r(1-b)_r.} \tag{3.6}$$

Proof. The case $r = 0$ is immediate. Suppose $r \geq 1$. Each D_j at $s = -r$ has total degree at most $j+1$ in a, b, with coefficients in $\mathbb{Q}[c]$. Every product contributing to A_r therefore has degree at most $2r$.

Fix an integer B with $1 \leq B \leq r$ and put $b = B$. The finite-difference identity $B_{j+1}(x+M) - B_{j+1}(x) = (j+1)\sum_{h=0}^{M-1}(x+h)^j$ gives

$$D_j(a,B,-r;c) = \frac{(-1)^{j+1}}{j}\left(\sum_{h=1}^{r-B}(a-c-h)^j + \sum_{h=1}^{B-1}(h-c)^j\right). \tag{3.7}$$

Exponentiating the formal logarithms yields the finite product

$$\sum_{m\geq 0} A_m(a,B,-r;c)t^m = \prod_{h=1}^{r-B}(1+(a-c-h)t)\prod_{h=1}^{B-1}(1+(h-c)t). \tag{3.8}$$

Its degree is at most $r-1$. Hence $A_r(a,B,-r;c)=0$. Symmetry in a,b supplies the roots $a=1,\dots,r$ as well. Thus $(1-a)_r(1-b)_r$ divides A_r in $\mathbb{Q}[c,a,b]$. The degree bound makes the quotient a polynomial in c alone.

Only $D_1^r/r!$ can contribute total degree $2r$: using any D_j with $j\geq 2$ makes the degree strictly smaller. The degree-two homogeneous part of $D_1(a,b,-r;c)$ in a,b is $-ab$. Therefore the coefficient of a^rb^r in A_r is $(-1)^r/r!$. The coefficient of a^rb^r in $(1-a)_r(1-b)_r$ is one. This determines the quotient and proves the identity, including its independence of c. □

Remark 3.3. The residue identity in [3] was already an analytic theorem. The proof above resolves the separate finite-algebraic proof request in [2]. The following section goes beyond that request by finding every coefficient in the resonant tail.

4 The entire resonant tail: Stirling formulas and an Appell family

Let $\left\{ {\ell \atop j} \right\}$ be a Stirling number of the second kind, with $\left\{ {0 \atop 0} \right\} = 1$. Define the finite polynomial

$$Q_{r,k}(a,b;c) = \sum_{\ell=0}^{k}\binom{k}{\ell}c^{k-\ell}\sum_{j=0}^{\ell}(-1)^j\left\{ {\ell \atop j} \right\}\frac{(a)_j(b)_j}{(r+1)_j}. \tag{4.1}$$

Because $r+1$ is a positive integer, there is no singular denominator.

Theorem 4.1 (Finite formula for every resonant coefficient)**.** *For all integers $r,k\geq 0$,*

$$\boxed{A_{r+k}(a,b,-r;c) = \frac{(-1)^r}{r!}(1-a)_r(1-b)_rQ_{r,k}(a,b;c).} \tag{4.2}$$

Equivalently, the normalized tail has the convergent local exponential generating function

$$\sum_{k\geq 0}Q_{r,k}(a,b;c)\frac{t^k}{k!} = \mathrm{e}^{ct}\,{}_2F_1(a,b;r+1;1-\mathrm{e}^t). \tag{4.3}$$

It satisfies

$$\partial_c Q_{r,k} = kQ_{r,k-1}, \qquad Q_{r,k}(c+h) = \sum_{j=0}^{k}\binom{k}{j}h^jQ_{r,k-j}(c). \tag{4.4}$$

Proof. First take $c=0$. The finite Gamma-ratio asymptotic expansion used in [3], or its formal Bernoulli derivation, identifies

$$\mathscr{W}(x) = x^{r-1}\sum_{j\geq 0}A_j(a,b,-r;0)x^{-j} \tag{4.5}$$

as the formal Laurent expansion of $\Gamma(x+a)\Gamma(x+b)/[\Gamma(x+1)\Gamma(x+a+b-r)]$. For parameters in a right half-plane this follows from the usual finite log-Gamma expansion; each coefficient identity then extends polynomially to all a,b. The Gamma recurrence consequently gives the formal identity

$$(x+1)(x+a+b-r)\mathscr{W}(x+1) = (x+a)(x+b)\mathscr{W}(x). \tag{4.6}$$

Only this formal recurrence is used in the rest of the proof.

Discard the polynomial part of $\mathscr{W}$ and send x^{-k-1} to $t^k/k!$. Write the resulting formal power series as

$$T(t) = \sum_{k\geq 0} A_{r+k}(a, b, -r; 0)\frac{t^k}{k!}. \tag{4.7}$$

Multiplication by x becomes ∂_t after polynomial parts are discarded; replacing x by $x+1$ becomes multiplication by e^{-t}. These statements follow directly on each monomial x^{-k-1}. Polynomial terms remain polynomial under both operations, so their omission does not affect the transformed identity. With $D = \partial_t$, (4.6) becomes

$$(D+1)(D+a+b-r)(\mathrm{e}^{-t}T) = (D+a)(D+b)T.$$

After simplification,

$$(1-\mathrm{e}^t)T'' + (a+b-r-1-(a+b)\mathrm{e}^t)T' - ab\mathrm{e}^t T = 0. \tag{4.8}$$

Set $z = 1-\mathrm{e}^t$. For $T(t) = F(z)$ this is Gauss's equation

$$z(1-z)F'' + [r+1-(a+b+1)z]F' - abF = 0. \tag{4.9}$$

The unique formal solution with constant term $F(0)$ is $F(0)_2F_1(a, b; r+1; z)$: comparison of successive coefficients divides only by $(j+1)(r+1+j)$, which is nonzero. Theorem 3.2 determines $F(0) = A_r$. This proves (4.3) at $c = 0$, with the residue prefactor restored.

At $s = -r$ and $m = r+k$, (3.5) has $\binom{s+m}{j} = \binom{k}{j}$, which vanishes for $j > k$. Thus comparison-shift transport multiplies the exponential generating function of the tail by e^{ct}. Finally, $(\mathrm{e}^t-1)^j = j!\sum_{\ell\geq j}\left\{{\ell \atop j}\right\} t^\ell/\ell!$ converts the hypergeometric series into (4.1). This proves every assertion. Local convergence follows, for example, in a sufficiently small disc where $|1-\mathrm{e}^t| < 1$. □

This is an exponential generating function for the normalized tail, not a claim of convergence of the original inverse-power expansion (4.5). The latter remains a formal asymptotic series. The explicit polynomials begin with

$$\begin{aligned} Q_{r,0} &= 1, \\ Q_{r,1} &= c - \frac{ab}{r+1}, \\ Q_{r,2} &= c^2 - \frac{2ab}{r+1}c + \frac{ab(ab+a+b-r-1)}{(r+1)(r+2)}. \end{aligned} \tag{4.10}$$

For $r = 1$ this gives

$$A_2(a, b, -1; c) = -(1-a)(1-b)\left(c - \frac{ab}{2}\right).$$

Unlike a recursive Bernoulli construction, (4.1) has a finite triangular formula whose summation ranges are immediately visible. It also gives the ordinary polynomial primitives $\int Q_{r,k}(c)\,\mathrm{d}c = Q_{r,k+1}(c)/(k+1) + \text{constant}$.

4.1 Integer-root collapse and nonsingular parameter jets

Equation (3.8) proves more than the vanishing of A_r: if one upper parameter lies in $\{1,\dots,r\}$, every A_m with $m \ge r$ vanishes at $s=-r$. The tail formula exhibits the full common factor responsible for this collapse.

Write $P_r(a) = (1-a)_r$. At an integer $A \in \{1,\dots,r\}$, the following analytic germ avoids logarithmic differentiation through a zero:

$$P_r(A+x) = (-1)^A (A-1)!(r-A)!\,x \times \exp\left[-\sum_{j\ge1} \frac{H^{(j)}_{r-A} + (-1)^j H^{(j)}_{A-1}}{j} x^j\right]. \tag{4.11}$$

To prove it, isolate the factor $-x$ in $\prod_{k=1}^{r}(k-A-x)$; the other factors are finite products of $1+x/j$ and $1-x/j$. Thus all mixed jets of (4.2) at integer roots are finite harmonic polynomials. For the residue itself,

$$\partial_a\partial_b A_r(a,b,-r;c)|_{a=A,b=B} = \frac{(-1)^{r+A+B}}{r!}(A-1)!(r-A)!(B-1)!(r-B)!. \tag{4.12}$$

No singular polygamma value is involved.

The normal, or s-direction, jets are different. For example,

$$\partial_c\partial_s^{\ell} A_r\Big|_{s=-r} = \ell\,\partial_s^{\ell-1} A_{r-1}\Big|_{s=-r} \qquad (\ell \ge 1). \tag{4.13}$$

Therefore comparison-shift independence of A_r on the resonant hyperplane does *not* imply independence of its normal derivatives. The distinction matters in Stieltjes coefficient extraction.

5 Stieltjes jets and an all-order shifted logarithmic identity

5.1 How the tail enters resonant Gauss summation

For $\Re a, \Re b, \Re c > 0$, set

$$W_n(a,b;s) = \frac{\Gamma(n+a)\Gamma(n+b)}{\Gamma(n+a+b+s)\Gamma(n+1)}.$$

The reciprocal Gamma in the denominator is understood as an entire function. The normally convergent, successively subtracted series

$$\mathcal{F}^{[M]}(s) = \sum_{n\ge0}\left[W_n(a,b;s) - \sum_{j=0}^{M} A_j(a,b,s;c)(n+c)^{-1-s-j}\right] \tag{5.1}$$

is holomorphic for $\Re s > -M-1$. Its exact value is

$$\mathcal{F}^{[M]}(s) = \frac{\Gamma(a)\Gamma(b)\Gamma(s)}{\Gamma(a+s)\Gamma(b+s)} - \sum_{j=0}^{M} A_j(a,b,s;c)\zeta(1+s+j,c). \tag{5.2}$$

This is the existing uniform subtraction theorem of [3]. For clarity, its justification is finite-order, locally uniform log-Gamma asymptotics followed by Gauss summation in $\Re s > 0$ and analytic continuation. The summand in (5.1) is $O(n^{-M-2-\Re s})$ uniformly on compact parameter sets.

Here is the precise coefficient rule, useful for applying the new tail formula without losing a Stieltjes term. Fix $r \le M$, and write

$$A_j(a,b,-r+\varepsilon;c) = \sum_{\ell\ge 0} a_{j,\ell}\varepsilon^\ell,$$
$$R(\varepsilon) = \varepsilon\frac{\Gamma(a)\Gamma(b)\Gamma(-r+\varepsilon)}{\Gamma(a-r+\varepsilon)\Gamma(b-r+\varepsilon)} = \sum_{\ell\ge 0} R_\ell\varepsilon^\ell.$$

The germ R is analytic, including at zeros of either reciprocal Gamma. For $k \ge 0$, direct Laurent multiplication gives

$$[\varepsilon^k]\mathcal{F}^{[M]}(-r+\varepsilon) = R_{k+1} - a_{r,k+1} - \sum_{\ell=0}^{k} a_{r,\ell}\frac{(-1)^{k-\ell}\gamma_{k-\ell}(c)}{(k-\ell)!} - \sum_{\substack{0\le j\le M\\ j\ne r}}\sum_{\ell=0}^{k} a_{j,\ell}\frac{\zeta^{(k-\ell)}(1-r+j,c)}{(k-\ell)!}. \tag{5.3}$$

The superscript on ζ denotes spectral differentiation. Theorem 4.1 now gives every $a_{r+j,0}$ by a finite Stirling sum. The coefficients $a_{r+j,\ell}$ for $\ell > 0$ still require normal parameter differentiation; replacing them by tangential derivatives of the resonant formula would be incorrect.

5.2 A shifted Beta–Stieltjes generating theorem

The following especially simple specialization gives ordinary convergent identities at every logarithmic order. Define

$$P_\ell(x) = [\varepsilon^\ell]\frac{\Gamma(x)}{\Gamma(x+\varepsilon)} = \frac{1}{\ell!}Y_\ell(-\psi(x),-\psi'(x),\dots,-\psi^{(\ell-1)}(x)), \tag{5.4}$$

where Y_ℓ is the complete exponential Bell polynomial and $P_0 = 1$. Equivalently, the first equality defines the polynomials without Bell notation. Set $b_{-1}(x) = 1$, $b_j(x) = (-1)^j\gamma_j(x)/j!$ for $j \ge 0$, and $b_j(x) = 0$ for $j < -1$.

Theorem 5.1 (All shifted logarithmic polygamma sums)**.** *For $\Re x > 0$, the series*

$$\mathcal{D}_x(\varepsilon) = \sum_{n\ge 0}\left[\frac{\Gamma(n+x)}{\Gamma(n+x+\varepsilon)} - (n+x)^{-\varepsilon} - \frac{\varepsilon(1-\varepsilon)}{2}(n+x)^{-1-\varepsilon}\right] \tag{5.5}$$

is normally convergent for $\Re\varepsilon > -1$. It equals the holomorphic completion of

$$\boxed{\mathcal{D}_x(\varepsilon) = \frac{\Gamma(x)}{(\varepsilon-1)\Gamma(x+\varepsilon-1)} - \zeta(\varepsilon,x) - \frac{\varepsilon(1-\varepsilon)}{2}\zeta(1+\varepsilon,x).} \tag{5.6}$$

For every integer $\ell \ge 1$, its coefficient identity is

$$\sum_{n\ge 0}\left[P_\ell(n+x) - \frac{(-\log(n+x))^\ell}{\ell!} - \frac{1}{2(n+x)}\left\{\frac{(-\log(n+x))^{\ell-1}}{(\ell-1)!} - \mathbf{1}_{\ell\ge 2}\frac{(-\log(n+x))^{\ell-2}}{(\ell-2)!}\right\}\right]$$
$$= P_\ell(x) - x\sum_{j=0}^{\ell} P_j(x) - \frac{\zeta^{(\ell)}(0,x)}{\ell!} - \frac{1}{2}b_{\ell-1}(x) + \frac{1}{2}b_{\ell-2}(x). \tag{5.7}$$

The grouped series in this formula is absolutely convergent.

Proof. The uniform two-term Gamma-ratio expansion is

$$\frac{\Gamma(y)}{\Gamma(y+\varepsilon)} = y^{-\varepsilon}\left(1 + \frac{\varepsilon(1-\varepsilon)}{2y} + O(y^{-2})\right), \qquad y = n + x \longrightarrow \infty.$$

Uniformity on compact parameter sets proves normal convergence of (5.5); all fixed parameter derivatives remain normally convergent. In $\Re\varepsilon > 1$, the beta integral gives

$$\begin{aligned}\sum_{n\geq 0} \frac{\Gamma(n+x)}{\Gamma(n+x+\varepsilon)} &= \frac{1}{\Gamma(\varepsilon)} \int_0^1 t^{x-1}(1-t)^{\varepsilon-1} \sum_{n\geq 0} t^n \, \mathrm{d}t \\ &= \frac{\Gamma(x)}{(\varepsilon-1)\Gamma(x+\varepsilon-1)}.\end{aligned}$$

The two comparison series sum to the displayed Hurwitz terms. Analytic continuation proves (5.6) on the larger half-plane, including the removable singularities at $\varepsilon = 0, 1$.

For coefficient extraction, the Gamma recurrence rewrites the first term of (5.6) as

$$\left(1 - \frac{x}{1-\varepsilon}\right) \frac{\Gamma(x)}{\Gamma(x+\varepsilon)}.$$

Its coefficient of ε^ℓ is $P_\ell(x) - x\sum_{j=0}^{\ell} P_j(x)$. Multiplication by $\varepsilon(1-\varepsilon)/2$ in the Hurwitz Laurent expansion gives $(b_{\ell-1} - b_{\ell-2})/2$. The remaining terms are ordinary Taylor coefficients. On the series side, every grouped coefficient is $O(n^{-2}(1+\log^\ell n))$ on compact x-sets; hence extraction commutes with summation and proves (5.7). □

These formulas are consequences of a classical beta summation and precise subtraction, not claims of new transcendental constants. Their usefulness is an explicit, all-order normalization connecting nonlinear harmonic polynomials to Stieltjes data and spectral derivatives.

5.3 Two explicit identities

Using $\zeta'(0,x) = \log\Gamma(x) - \frac{1}{2}\log(2\pi)$, the first order is

$$\begin{aligned}&\sum_{n\geq 0}\left[\log(n+x) - \psi(n+x) - \frac{1}{2(n+x)}\right] \\ &\quad = \left(x - \frac{1}{2}\right)\psi(x) - \log\Gamma(x) - x + \frac{1}{2} + \frac{1}{2}\log(2\pi). \qquad (5.8)\end{aligned}$$

The second order is

$$\begin{aligned}&\sum_{n\geq 0}\left[\psi(n+x)^2 - \psi'(n+x) - \log^2(n+x) + \frac{1+\log(n+x)}{n+x}\right] \\ &\quad = (1-x)(\psi(x)^2 - \psi'(x)) + (2x-1)\psi(x) - 2x + \gamma_1(x) - \zeta''(0,x). \qquad (5.9)\end{aligned}$$

At $x = 1$, these become

$$\sum_{k\geq 1}\left[\log k - H_{k-1} + \gamma - \frac{1}{2k}\right] = \frac{\log(2\pi) - 1 - \gamma}{2}, \qquad (5.10)$$

$$\sum_{k\geq 1}\left[(H_{k-1} - \gamma)^2 + H_{k-1}^{(2)} - \zeta(2) - \log^2 k + \frac{1+\log k}{k}\right] = -2 - \gamma + \gamma_1 - \zeta''(0). \qquad (5.11)$$

The grouping is essential. These are not identities obtained by separately summing their generally divergent components.

6 Moving reflection, common zeros, and unique centering

The local identity underlying the reflection theorem is

$$c_j(u, 1+u-a) = (-1)^j c_j(u, a). \tag{6.1}$$

Indeed, $\phi(t, u, 1+u-a) = \phi(-t, u, a)$ as analytic germs. One can also see it immediately from (2.7). The term u in the reflected shift cannot be dropped.

Theorem 6.1 (Moving-reflection zeros at every depth)**.** *Let $0 < \Re a < 1$ and take u in a sufficiently small neighbourhood of zero. Define*

$$\mathcal{R}_\pm(s, u; a) = \mathcal{B}(s, u; a) \pm \mathcal{B}(s, u; 1+u-a).$$

For every $m \geq 0$, $\mathcal{R}_+$ is jointly holomorphic near $(s, u) = (-2m, 0)$ and

$$\mathcal{R}_+(-2m, u; a) = 0. \tag{6.2}$$

Similarly, $\mathcal{R}_-$ is jointly holomorphic near $(-2m-1, 0)$ and vanishes at $s = -2m-1$. Its depth coefficients are the explicit finite combinations

$$[u^r]\mathcal{R}_\pm(s, u; a) = Z_r(s, a) \pm \sum_{k=0}^{r} \frac{(-1)^k (s)_k}{k!} Z_{r-k}(s+k, 1-a). \tag{6.3}$$

Every coefficient inherits the stated holomorphic zero.

Proof. Apply Lemma 2.1 to both factors with the reflected shift inserted before coefficient extraction. By (6.1), the plus combination has only even heat indices j, while the minus combination has only odd indices. Near $s = -2m, u = 0$, the only potentially vanishing denominator in (2.6) has $j = 2m+1$; its coefficient in the plus combination is identically zero in u. Thus $\Gamma(s)\mathcal{R}_+$ is jointly holomorphic there. The reciprocal Gamma has a zero at $s = -2m$, proving the holomorphy and the zero of $\mathcal{R}_+$ itself. The minus case uses the excluded even index $j = 2m+2$ in exactly the same argument.

For the coefficient formula, Taylor-expand the shift: $\partial_a^k \mathcal{B}(s, u; a) = (-1)^k (s)_k \mathcal{B}(s+k, u; a)$ in the ordinary convergence half-plane. Taking coefficients gives (6.3); meromorphic continuation extends the identity. Joint holomorphy at the indicated points permits coefficient extraction there and transfers the zero to every depth. □

6.1 The centered family

Put

$$\mathcal{C}(s, u) = \mathcal{B}\left(s, u; \frac{1+u}{2}\right), \qquad C_r(s) = [u^r]\mathcal{C}(s, u). \tag{6.4}$$

Its Mellin kernel is particularly simple:

$$\Gamma(s)\mathcal{C}(s, u) = \int_0^\infty t^{s-1} \left(2\sinh(t/2)\right)^{-1-u} \mathrm{d}t. \tag{6.5}$$

The heat factor $(t/(2\sinh(t/2)))^{1+u}$ is even.

Corollary 6.2 (Centered all-depth trivial zeros)**.** *For all integers $r, m \geq 0$, the meromorphic combination*

$$\boxed{C_r(s) = \sum_{k=0}^{r} \frac{(-1)^k (s)_k}{2^k k!} Z_{r-k}(s+k, 1/2)} \tag{6.6}$$

is holomorphic at $s = -2m$ and satisfies

$$\boxed{C_r(-2m) = 0.} \tag{6.7}$$

Proof. The even heat factor gives the same missing-pole argument as Theorem 6.1. Taylor expansion of the shift by $u/2$ gives (6.6). □

The identity is about the continued *combination*. Individual terms on its right can have poles, and they must not be evaluated separately at a singular point. At depth zero it reduces to the classical identity $C_0(s) = \zeta(s, 1/2) = (2^s - 1)\zeta(s)$ and its even zeros. At depth one,

$$C_1(s) = Z_1(s, 1/2) - \frac{s}{2}\zeta(s+1, 1/2). \tag{6.8}$$

Consequently $Z_1(0, 1/2) = 1/2$, and for $m \geq 1$,

$$Z_1(-2m, 1/2) = -m\zeta(1-2m, 1/2) = \frac{B_{2m}(1/2)}{2}. \tag{6.9}$$

Here the individual Z_1 values are holomorphic.

At depth two and $m \geq 1$, expansion of (6.6) gives

$$\operatorname{Res}_{s=-2m} Z_2(s, 1/2) = -m\zeta(1-2m, 1/2), \tag{6.10}$$

$$\operatorname{FP}_{s=-2m} Z_2(s, 1/2) = -m \operatorname{FP}_{s=1-2m} Z_1(s, 1/2) + \tfrac{1}{2}\zeta(1-2m, 1/2). \tag{6.11}$$

The last $\frac{1}{2}\zeta$ term comes from the first-order variation of $-s/2$ multiplying a pole. The remaining Hurwitz term is zero because $\zeta(2-2m, 1/2) = 0$. For example,

$$Z_2(-2+\varepsilon, 1/2) = -\frac{1}{24\varepsilon} + \frac{1-2\gamma}{48} + O(\varepsilon). \tag{6.12}$$

The local coefficient rule of [4], reproduced briefly in Section 11, supplies these individual Laurent data.

6.2 Why the center is forced

Theorem 6.3 (Unique centering; two-depth rigidity for affine shifts)**.** *Let $a(u)$ be analytic near zero with $\Re a(0) > 0$. Every coefficient of $\mathcal{B}(s, u; a(u))$ in u is holomorphic and zero at $s = 0$ if and only if*

$$a(u) = \frac{1+u}{2} \tag{6.13}$$

as an analytic germ. In the affine class $a(u) = a_0 + \lambda u$, it is enough to require that the depth-zero and depth-one coefficients be holomorphic and zero at $s = 0$.

Proof. The resonant heat coefficient at $s = 0$ is $q(u) = c_1(u, a(u)) = (1+u)/2 - a(u)$. Lemma 2.1 gives

$$\mathcal{B}(s, u; a(u)) = \frac{1}{\Gamma(s)} \left(\frac{q(u)}{s-u} + Q(s, u) \right),$$

with Q jointly holomorphic near $(0,0)$. Since $1/\Gamma(s) = s + O(s^2)$, the Q term is $O(s)$ after any fixed depth extraction. Suppose the first nonzero coefficient of q is $q_j u^j$. The depth-j coefficient of the displayed singular term then has constant term q_j at $s = 0$, contradicting the hypothesis. Thus $q = 0$. The converse is Corollary 6.2.

For an affine shift, $q(u) = 1/2 - a_0 + (1/2 - \lambda)u$. The depth-zero condition forces $a_0 = 1/2$; the depth-one condition then forces $\lambda = 1/2$. All higher even zeros follow automatically. □

This is a finite rigidity phenomenon: in the affine class, two local conditions select a family with infinitely many zero identities at all harmonic depths.

7 An all-depth Gamma evaluation of ordinary harmonic sums

Let $L = \log 2$, and introduce the centered Gamma quotient

$$F_c(u) = \frac{\Gamma((1+u)/2)\Gamma(1-u)}{\Gamma((1-u)/2)}. \tag{7.1}$$

The subscript denotes "centered", not the comparison shift of the Gauss coefficients.

Theorem 7.1 (Centered weight-two Gamma generating function)**.** *For $|u| < 1$, with the removable value at $u = 0$ understood,*

$$\boxed{\mathcal{C}(2,u) = \frac{\pi \tan(\pi u/2)}{u} F_c(u).} \tag{7.2}$$

Equivalently,

$$\mathcal{C}(2,u) = \frac{\pi^2}{2} \exp\left(\sum_{j \geq 1} \frac{\kappa_j}{j} u^j \right), \tag{7.3}$$

where

$$\kappa_1 = -2L, \qquad \kappa_j = \begin{cases} (3 - 2^{2-j})\zeta(j), & j \geq 2 \text{ even}, \\ (2^{1-j} - 1)\zeta(j), & j \geq 3 \text{ odd}. \end{cases} \tag{7.4}$$

If $h_0 = 1$ and $r h_r = \sum_{j=1}^{r} \kappa_j h_{r-j}$, then $C_r(2) = \pi^2 h_r/2$ for every $r \geq 0$.

Proof. For $\Re u < 0$, beta integration at $s = 1$ gives $\mathcal{B}(1,u;a) = B(a,-u)$. Differentiating in a gives

$$\mathcal{B}(2,u;a) = -B(a,-u)\big(\psi(a) - \psi(a-u)\big).$$

At $a = (1+u)/2$, one has $a - u = 1 - a$, and the digamma reflection formula implies $\psi(a) - \psi(1-a) = -\pi \cot(\pi a) = \pi \tan(\pi u/2)$. Also $B(a,-u) = -F_c(u)/u$. This proves (7.2) initially in a left half-disc. Both sides are analytic in $|u| < 1$ after removing the value at zero: on the series side the summands are $O(n^{-2+\Re u})$, locally uniformly, and the moving shift retains positive real part. Analytic continuation proves the stated domain.

Taylor expansion of the three log-Gamma factors gives

$$\log F_c(u) = -2Lu + \sum_{\substack{j \geq 2 \\ j \text{ even}}} \frac{\zeta(j)}{j} u^j + \sum_{\substack{j \geq 3 \\ j \text{ odd}}} \frac{(2^{1-j} - 1)\zeta(j)}{j} u^j. \tag{7.5}$$

Here the odd Taylor coefficients at $1/2$ use $\psi^{(j-1)}(1/2) = (-1)^j(j-1)!(2^j-1)\zeta(j)$ for $j \geq 2$; the linear term uses $\psi(1/2) = -\gamma - 2L$. The sine product, or its Taylor series, yields

$$\log \frac{\tan(\pi u/2)}{\pi u/2} = \sum_{k\geq 1} \frac{(1-2^{1-2k})\zeta(2k)}{k} u^{2k}.$$

Adding this to (7.5) gives (7.4). Coefficient extraction from the exponential gives the finite recursion. □

7.1 Explicit generalized harmonic identities

The coefficient of $(1+u/(2n+1))^{-2}$ gives the following ordinary series interpretation of every h_r:

$$\boxed{\sum_{n\geq 0} \frac{1}{(2n+1)^2} \sum_{j=0}^{r} \frac{(-1)^j(j+1)E_{n,r-j}}{(2n+1)^j} = \frac{\pi^2}{8} h_r.} \tag{7.6}$$

The $n=0$ terms are included. The bound $E_{n,r} = O_r((1+\log(n+1))^r)$ proves absolute convergence for each fixed depth. This is a family of identities, not an assertion that an arbitrary nonlinear harmonic numerator has the same reduction.

The first coefficients are

$$\begin{aligned} h_0 &= 1, & h_1 &= -2L, \\ h_2 &= 2L^2 + \zeta(2), & h_3 &= -\frac{4}{3}L^3 - 2L\zeta(2) - \frac{1}{4}\zeta(3), \\ h_4 &= \frac{2}{3}L^4 + 2L^2\zeta(2) + \frac{1}{2}L\zeta(3) + \frac{1}{2}\zeta(2)^2 + \frac{11}{16}\zeta(4). \end{aligned} \tag{7.7}$$

For example, writing $q_n = 2n+1$,

$$\sum_{n\geq 0} \frac{H_n - 2/q_n}{q_n^2} = -\frac{\pi^2}{4} L, \tag{7.8}$$

$$\sum_{n\geq 0} \frac{H_n^2 - H_n^{(2)} - 4H_n/q_n + 6/q_n^2}{q_n^2} = \frac{\pi^2}{4}(2L^2 + \zeta(2)), \tag{7.9}$$

$$\sum_{n\geq 0} \frac{E_{n,3} - 2E_{n,2}/q_n + 3H_n/q_n^2 - 4/q_n^3}{q_n^2} = \frac{\pi^2}{8}\left(-\frac{4}{3}L^3 - 2L\zeta(2) - \frac{1}{4}\zeta(3)\right). \tag{7.10}$$

The complete recurrence, rather than only these small cases, is the content of (7.6). The data file `exact_tables.json` supplies coefficients through depth eight.

There is a second, independent integral interpretation:

$$\frac{1}{r!}\int_0^\infty \frac{t\,[-\log(2\sinh(t/2))]^r}{2\sinh(t/2)}\,\mathrm{d}t = \frac{\pi^2}{2} h_r. \tag{7.11}$$

At zero the integrand is $O(1+|\log t|^r)$; at infinity it decays exponentially times a polynomial. Thus this is an ordinary integral. Differentiation of (6.5) at $s=2$ is justified by these majorants. These moments can change sign; no positivity claim is being used.

7.2 All positive outer exponents

Let

$$F(a,u)=\frac{\Gamma(a)\Gamma(1-u)}{\Gamma(a-u)},\qquad \mathcal{C}(s,u;x)=\mathcal{B}\left(s,u;\frac{1+u}{2}+x\right).$$

For every integer $p\geq 2$ and $|\Re x|<1/2$, one obtains

$$[u^r]\mathcal{C}(p,u;x)=\frac{(-1)^p}{(p-1)!}[u^{r+1}]\,\partial_a^{p-1}F(a,u)\Big|_{a=(1+u)/2+x}. \tag{7.12}$$

Indeed, $\mathcal{B}(p,u;a)=(-1)^{p-1}\partial_a^{p-1}B(a,-u)/(p-1)!$; substitute $B(a,-u)=-F(a,u)/u$ and extract the coefficient. The beta identity first holds for $\Re u<0$ and then extends through zero after differentiation in a. Uniform convergence of the differentiated series for $|u|$ small supplies the same conclusion directly.

For $x=0$ every coefficient in (7.12) is a finite polynomial in L and zeta values at positive integers. Formula (7.2) is the particularly simple reflection collapse at $p=2$; the other exponents retain finite polygamma/Bell expressions. This general positive-integer beta mechanism is classical and overlaps the fixed-shift theory in [4].

8 Centered Stieltjes constants and polylogarithmic coordinates

Write

$$G(u)=\frac{1}{\Gamma(1+u)}=\sum_{j\geq 0}g_j u^j,\qquad F_c(u)=\sum_{j\geq 0}f_j u^j. \tag{8.1}$$

Thus $g_0=1$, $g_1=\gamma$, and $g_2=(\gamma^2-\zeta(2))/2$.

Theorem 8.1 (Centered principal part and finite part)**.** *For every $r\geq 0$, as $\varepsilon\to 0$,*

$$\boxed{C_r(1+\varepsilon)=\sum_{j=0}^{r}\frac{g_j}{\varepsilon^{r+1-j}}+g_{r+1}-f_{r+1}+O(\varepsilon).} \tag{8.2}$$

In particular, the finite part is an explicit polynomial in $\gamma, L, \zeta(2),\dots,\zeta(r+1)$.

Proof. The only moving pole near $(s,u)=(1,0)$ has residue $1/\Gamma(1+u)=G(u)$, because the leading heat coefficient is one. Consequently

$$\mathcal{C}(s,u)=\frac{G(u)}{s-1-u}+Q(s,u)$$

with Q jointly holomorphic. This follows from Lemma 2.1, replacing the numerator $1/\Gamma(s)$ by $1/\Gamma(1+u)$; their difference divided by $s-1-u$ is holomorphic. For $\Re u<0$, beta integration gives $\mathcal{C}(1,u)=-F_c(u)/u$, and therefore $Q(1,u)=(G(u)-F_c(u))/u$. Both sides extend through $u=0$. Expanding the simple moving pole for $|u|<|\varepsilon|$ and then continuing the coefficient functions proves (8.2). □

For the first three depths,

$$\begin{aligned}
&\mathrm{FP}_{s=1}\,C_0(s)=\gamma+2L,\\
&\mathrm{FP}_{s=1}\,C_1(s)=\frac{\gamma^2}{2}-\zeta(2)-2L^2,\\
&\mathrm{FP}_{s=1}\,C_2(s)=g_3+\frac{4}{3}L^3+L\zeta(2)+\frac{1}{4}\zeta(3).
\end{aligned} \tag{8.3}$$

These are finite parts of the centered family, not the same constants as those for a fixed outer shift $1/2$ at positive harmonic depth.

8.1 Every higher Laurent coefficient: exact transport, not false closure

Define fixed-shift harmonic Stieltjes coefficients by

$$Z_r(1+\varepsilon,a)=\sum_{j=0}^{r} g_j\varepsilon^{-r-1+j}+\sum_{\ell\geq 0}\frac{(-1)^\ell\gamma_{r,\ell}(a)}{\ell!}\varepsilon^\ell. \tag{8.4}$$

Define $\widetilde{\gamma}_{r,\ell}$ in the same way for C_r. At depth zero these are the usual $\gamma_\ell(1/2)$. For $e_j(k)=[v^j]\prod_{h=1}^{k}(1+v/h)$, expansion of (6.6) gives the exact finite transport law

$$\widetilde{\gamma}_{r,\ell}=\gamma_{r,\ell}(1/2)+(-1)^\ell\ell!\sum_{k=1}^{r}\frac{(-1)^k}{2^k}\sum_{j=0}^{\ell}e_j(k)\frac{\partial_s^{\ell-j}Z_{r-k}(k+1,1/2)}{(\ell-j)!}. \tag{8.5}$$

Here $e_j(k)=0$ for $j>k$. To prove the formula, observe that $(1+\varepsilon)_k/k!=\sum_j e_j(k)\varepsilon^j$ and that every $Z_{r-k}(k+1+\varepsilon,1/2)$ with $k\geq 1$ is holomorphic. The principal part is unchanged, and ordinary Taylor multiplication proves the identity.

For $\ell=0$, all terms reduce by beta differentiation and reproduce Theorem 8.1. For $\ell>0$, genuine spectral derivatives remain. Neither centering nor the common zeros justify discarding them. This is precisely the separation between local finite parts and higher harmonic Stieltjes data required in the source theory.

The ordinary depth-zero constants satisfy the familiar half-shift law

$$\gamma_\ell(1/2)=\gamma_\ell+\frac{2(-1)^\ell L^{\ell+1}}{\ell+1}+2\sum_{j=0}^{\ell-1}\binom{\ell}{j}(-1)^{\ell+j}L^{\ell-j}\gamma_j. \tag{8.6}$$

This follows directly by expanding $\zeta(s,1/2)=(2^s-1)\zeta(s)$ at $s=1$.

8.2 Exact polylogarithmic coordinates at rational shifts

For $x=p/q$, $1\leq p\leq q$, and $\omega=\mathrm{e}^{2\pi i/q}$, finite character filtering gives

$$\zeta(s,p/q)=q^{s-1}\sum_{j=0}^{q-1}\omega^{-pj}\,\mathrm{Li}_s(\omega^j). \tag{8.7}$$

It is first proved by absolutely convergent series in $\Re s>1$ and then continued meromorphically. In the $j=0$ term, $\mathrm{Li}_s(1)$ means $\zeta(s)$; the other root-of-unity values use their radial continuation. At $s=0$ there is no pole, and every spectral derivative has the finite formula

$$\zeta^{(\ell)}(0,p/q)=\frac{1}{q}\sum_{j=0}^{q-1}\omega^{-pj}\sum_{h=0}^{\ell}\binom{\ell}{h}(\log q)^{\ell-h}\,\partial_s^h\,\mathrm{Li}_s(\omega^j)\Big|_{s=0}. \tag{8.8}$$

Together with the standard polygamma–Hurwitz relation $\psi^{(m)}(x)=(-1)^{m+1}m!\zeta(m+1,x)$ for $m\geq 1$, this converts Theorem 5.1 at positive rational shifts into exact identities among harmonic-polygamma sums, generalized Stieltjes constants, and order derivatives of finitely many polylogarithms. Integer translation handles rational shifts outside $(0,1]$.

The Mellin kernel itself also has a multiple-polylogarithm interpretation:

$$\sum_{n\geq 0} E_{n,r} z^n = \frac{[-\log(1-z)]^r}{r!(1-z)} = \frac{\mathrm{d}}{\mathrm{d}z} \operatorname{Li}_{\{1\}^{r+1}}(z), \qquad |z| < 1. \tag{8.9}$$

Thus the harmonic identities arise from Mellin transforms of actual polylogarithmic derivatives, rather than from an analogy of notation. Equation (8.7) evaluates a prescribed finite average; it does not imply reduction of its individual higher-depth analogues.

9 Ordinary antiderivatives and Gamma reflection integrals

9.1 A log-Gamma and Hurwitz primitive

Let $S_1(x)$ denote the ordinary sum in (5.8), for real $x > 0$. An explicit antiderivative is

$$\boxed{\int S_1(x)\,\mathrm{d}x = \left(x - \frac{1}{2}\right)\log\Gamma(x) - 2\zeta'(-1,x) + \frac{x^2-x}{2} - \frac{x}{2}\log(2\pi) + \text{constant}.} \tag{9.1}$$

Indeed, differentiating the Hurwitz shift identity in the spectral variable gives

$$\partial_x \zeta'(-1,x) = -\zeta(0,x) + \zeta'(0,x) = x - \tfrac{1}{2} + \log\Gamma(x) - \tfrac{1}{2}\log(2\pi).$$

Direct differentiation of (9.1) therefore gives (5.8). Normal convergence of the grouped series locally in $x > 0$ also permits ordinary integration term by term over a compact interval. The formula is not a finite-part integral.

9.2 Moving-centered reflection primitives at every depth

For $c_u = (1+u)/2$, let

$$\mathcal{C}_\pm(s,u;x) = \mathcal{B}(s,u;c_u + x) \pm \mathcal{B}(s,u;c_u - x).$$

In the initial convergence domain, differentiation gives

$$\partial_x \mathcal{C}_\pm(s,u;x) = -s\mathcal{C}_\mp(s+1,u;x). \tag{9.2}$$

It extends meromorphically and, for $s \neq 1$, supplies the primitive $-\mathcal{C}_\mp(s-1,u;x)/(s-1)$. At $s = 1$ this expression must be continued with its pole terms; there is no license to substitute into an indeterminate quotient.

At $s = 2$ the anchored primitives are ordinary integrals. For real $|x| < 1/2$ and u sufficiently small,

$$\int_0^x \mathcal{C}_+(2,u;v)\,\mathrm{d}v = B(c_u - x, -u) - B(c_u + x, -u), \tag{9.3}$$

$$\int_0^x \mathcal{C}_-(2,u;v)\,\mathrm{d}v = 2B(c_u, -u) - B(c_u + x, -u) - B(c_u - x, -u). \tag{9.4}$$

The apparent beta poles at $u = 0$ cancel. These identities follow by integrating $\mathcal{B}(2,u;a) = -\partial_a B(a,-u)$ first for $\Re u < 0$; normal convergence in a neighbourhood of $u = 0$ proves their extension. All depth coefficients give finite Gamma/polygamma primitives. For example, the depth-zero plus identity reduces to

$$\int_0^x \left[\psi'(1/2+v) + \psi'(1/2-v)\right]\mathrm{d}v = \pi\tan(\pi x). \tag{9.5}$$

There is also a useful fixed-a presentation of the moving endpoint:

$$\int_a^{1+u-a} \mathcal{B}(2,u;x)\,\mathrm{d}x = B(a,-u) - B(1+u-a,-u) \\ = \frac{\Gamma(a)\Gamma(-u)}{\Gamma(a-u)}\left(1 - \frac{\sin \pi a}{\sin \pi(a-u)}\right). \tag{9.6}$$

The ratio of the beta terms follows from Gamma reflection. For real $0 < a < 1$ and small real u, the oriented integral is ordinary; the complex-parameter statement is its analytic germ. At $u = 0$ the right side has a removable value $\pi \cot(\pi a)$. Higher coefficients include the moving-endpoint contributions automatically.

10 Several unequal frequencies and independent harmonic depths

The parity mechanism is not restricted to one summation variable or a single harmonic-depth parameter. Fix $d \geq 1$ and positive real frequencies $\omega_1, \ldots, \omega_d$. Put

$$\alpha_i = 1 + u_i, \qquad U = \sum_i u_i, \qquad a(\boldsymbol{u}) = \frac{1}{2}\sum_i \alpha_i \omega_i.$$

Define, initially in an absolute-convergence domain,

$$\mathcal{B}_{\boldsymbol{\omega}}(s,\boldsymbol{u}) = \sum_{n_1,\ldots,n_d \geq 0} \frac{\prod_{i=1}^d (1+u_i)_{n_i}/n_i!}{\left(a(\boldsymbol{u}) + \sum_i \omega_i n_i\right)^s}. \tag{10.1}$$

For instance, $\Re s > d + \sum_i |u_i|$ and all u_i sufficiently small provide a common domain. The denominator then has positive real part.

Theorem 10.1 (Unequal-frequency all-depth reflection zeros)**.** *Let $m \geq 0$ be an integer with $d + m$ odd. The family (10.1) is jointly holomorphic near $(s,\boldsymbol{u}) = (-m,\boldsymbol{0})$, and*

$$\mathcal{B}_{\boldsymbol{\omega}}(-m,\boldsymbol{u}) = 0. \tag{10.2}$$

Therefore every multi-depth coefficient $[\boldsymbol{u}^{\boldsymbol{r}}]$ has a holomorphic zero there. No rational dependence among the frequencies is required.

Proof. The product of binomial series yields

$$\Gamma(s)\mathcal{B}_{\boldsymbol{\omega}}(s,\boldsymbol{u}) = \int_0^\infty t^{s-1} \prod_{i=1}^d \left(2\sinh(\omega_i t/2)\right)^{-1-u_i}\,\mathrm{d}t. \tag{10.3}$$

Near zero, the kernel is

$$t^{-d-U}\prod_i \omega_i^{-1-u_i} \prod_i \left(\frac{\omega_i t}{2\sinh(\omega_i t/2)}\right)^{1+u_i}.$$

The last product is an even analytic function of t, jointly analytic in $\boldsymbol{u}$. Uniform Mellin subtraction therefore has possible moving poles only at $s = d + U - 2j$. At $s = -m, \boldsymbol{u} = 0$, a pole would require $2j = d + m$, excluded by the parity assumption. Choosing a sufficiently large subtraction order proves joint holomorphy of the Mellin side near the specified point. Multiplication by $1/\Gamma(s)$ then gives the zero. Uniform local holomorphy permits every multi-depth extraction. □

For $d = 1$ the protected points are $0, -2, -4, \ldots$; for $d = 2$ they are $-1, -3, -5, \ldots$. This is a statement about the family and all its order derivatives, not just about a fixed integer-order Barnes zeta.

To display the harmonic content explicitly, let $a_0 = \frac{1}{2}\sum_i \omega_i$ and, for a multi-index $\boldsymbol{r}$, define

$$Z^{\omega}_{\boldsymbol{r}}(s) = \sum_{\boldsymbol{n}\geq 0} \frac{\prod_i E_{n_i,r_i}}{(a_0 + \sum_i \omega_i n_i)^s}.$$

Taylor expansion of the moving center gives the finite exact formula

$$[\boldsymbol{u}^{\boldsymbol{r}}]\mathcal{B}_{\boldsymbol{\omega}}(s,\boldsymbol{u}) = \sum_{\boldsymbol{k}\leq \boldsymbol{r}} (-1)^{|\boldsymbol{k}|}(s)_{|\boldsymbol{k}|} \prod_i \frac{(\omega_i/2)^{k_i}}{k_i!} Z^{\boldsymbol{\omega}}_{\boldsymbol{r}-\boldsymbol{k}}(s+|\boldsymbol{k}|). \tag{10.4}$$

Theorem 10.1 gives all-order cancellation identities for these combinations. At a singular specialization, the complete combination is continued before its value is taken.

If the center is instead displaced by a fixed x, reflection $x \mapsto -x$ again separates the even and odd heat coefficients. The plus combination has the parity class of Theorem 10.1; the minus combination has the opposite class of nonpositive integers. This follows by multiplying the even heat factor by $2\cosh(xt)$ or $-2\sinh(xt)$, respectively.

The result does not say that all unequal-frequency coefficients reduce to ordinary zeta constants. For noncommensurate frequencies, the underlying Barnes-type periods remain distinct coordinates. The exact zero statement is independent of such a reduction problem.

11 Audit, necessary corrections to extensions, and verification

11.1 The fixed-shift negative Laurent theorem is prior work

The preserved source [4] already proves the following local rule (its outer-shift notation differs by one). With our heat coefficients and $m \geq 0$, put

$$\frac{c_{m+1}(u,a)}{\Gamma(u-m)} = \sum_{j\geq 1} \lambda_{m,j}(a) u^j.$$

Then

$$Z_r(-m+\varepsilon, a) = \sum_{h=0}^{r} \lambda_{m,r+1-h}(a)\varepsilon^{-h} + O(\varepsilon). \tag{11.1}$$

A short explanation is worth retaining: near $s = -m$, only the heat coefficient c_{m+1} has a moving denominator $\varepsilon - u$. All other terms are holomorphic and, after multiplication by $1/\Gamma(-m+\varepsilon)$, are $O(\varepsilon)$ at every fixed depth. Multiplying the remaining numerator by the reciprocal Gamma series produces exactly (11.1).

This is not a new result of the present continuation. It is included to normalize examples and to make the additional moving-reflection condition transparent. The analytic remainder starts at $O(\varepsilon)$, so it can and generally does affect positive spectral derivatives. In particular, at depth zero these include $\zeta'(-m,a)$, and at $m = 0$ they already include $\log\Gamma(a)$.

11.2 A version-specific indexing correction in a cited preprint

The unnumbered Stieltjes-limit display at the bottom of page 1 of [7], specifically arXiv version 1, starts its sum at $n = 1$ while using the Hurwitz normalization with $n = 0$. The PDF page was

checked directly. For real $a > 0$, the corrected formula is

$$\gamma_m(a) = \lim_{N\to\infty} \left(\sum_{n=0}^{N} \frac{\log^m(n+a)}{n+a} - \frac{\log^{m+1}(N+a)}{m+1} \right). \tag{11.2}$$

If the lower index $n = 1$ is retained, the missing term $\log^m(a)/a$ must be added. Indeed, $\zeta(s, a+1) = \zeta(s, a) - a^{-s}$ gives

$$\gamma_m(a+1) = \gamma_m(a) - \frac{\log^m a}{a}.$$

The printed $n = 1$ limit is $\gamma_m(a+1)$, not $\gamma_m(a)$. Already at $m = 0, a = 1$, it becomes $\lim_{N\to\infty}(H_{N+1} - 1 - \log(N+1)) = \gamma - 1$. This identifies a specific off-by-one error; it is not an objection to that paper's other results, and no claim is made about later revisions. The convention (2.8) and all identities in this report use the corrected normalization.

11.3 Three invalid extensions, with exact corrections

No contradiction to the cited Gauss subtraction or fixed-shift Laurent theorems was found in the targeted audit. The following are explicit corrections to tempting extensions, not allegations that those sources made the errors.

Frozen reflection is not moving reflection. Replacing $1 + u - a$ by $1 - a$ leaves the first heat coefficient sum equal to u. As a concrete consequence,

$$Z_1(s, a) + Z_1(s, 1-a) = 1 + O(s) \qquad (s \to 0), \tag{11.3}$$

for $0 < \Re a < 1$. The sum is holomorphic but is not zero. Its poles cancel because their leading heat coefficients are opposite; its constant terms sum to one. The extra shift-derivative term in (6.3) is exactly what removes this defect.

Normal jets are not determined by a resonant restriction. For $a = b = 1$ and $s = -1 + \varepsilon$, the exact polynomial is

$$A_1(1, 1, -1+\varepsilon; c) = (c - \tfrac{1}{2})\varepsilon - \tfrac{1}{2}\varepsilon^2. \tag{11.4}$$

Although its resonant value is zero and independent of c, its first normal derivative is $c - 1/2$. Formula (5.3) therefore needs actual normal derivatives; the restriction (3.6) cannot replace them.

Finite-part extraction does not commute with multiplication by a varying coefficient without residue terms. For a Laurent series $V(\varepsilon)$, $\mathrm{FP}(\varepsilon V) = \operatorname{Res} V$, not zero in general. Equation (6.11) displays the resulting correction explicitly. Similarly, omitting the second subtraction in (5.5) destroys the normal convergence needed for its all-order derivative formulas.

11.4 Reproducible checks and their limits

The supplied verifier uses two independent coefficient constructions: Bernoulli recursions for A_j and finite Stirling sums for $Q_{r,k}$. The exact run contains 436 identities: 16 fully symbolic resonant-tail checks, 189 higher-order rational checks, 5 comparison-shift identities, 25 Appell identities, 154 finite-product collapse checks, 13 moving reflection checks, 10 centered parity checks, 15 unequal-frequency parity checks, and 9 Gamma-cumulant checks. Four negative controls reject specified corruptions of a tail sign, a reflection shift, a center slope, and a Stieltjes sign.

The numerical run uses 65 decimal digits and makes 30 comparisons. These include ordinary Mellin integrals versus the Gamma quotient, log-sinh moments versus the independent cumulant recursion, subtracted Gamma-ratio sums versus (5.6), direct Hurwitz differentiation of (9.1), and Cauchy coefficient extraction for the second Stieltjes jet. The largest reported normalized discrepancy is approximately 1.99×10^{-51}. The Gamma-ratio series accelerator uses finite asymptotic tails; these floating-point diagnostics are *not* rigorous interval enclosures.

The finite tests do not prove the arbitrary-index theorems. Those are proved in the preceding sections. Conversely, a passing finite test should not promote a different pending source claim to proved status. There is no proof-assistant formalization in this package.

12 Further research questions and concrete next targets

12.1 Normal spectral jets of the resonant tail

Theorem 4.1 gives a closed finite formula for the full tail on $s = -r$. Find a similarly explicit representation of $\partial_s^\ell A_{r+k}|_{s=-r}$ for arbitrary ℓ. A useful first result would be a hypergeometric inhomogeneous differential equation for the Borel transforms at $\ell = 1, 2$, together with normalized initial data. Equation (4.13) is a necessary compatibility condition. A solution would make the Stieltjes identities (5.3) substantially more explicit beyond their constant coefficient.

12.2 Derivatives at the protected common zeros

The common zero $C_r(-2m) = 0$ is local and exact. Its derivative generally retains the analytic Mellin remainder. Determine which finite combinations of $C_r'(-2m)$ eliminate that remainder, and identify the natural coordinates for the first surviving cases, beginning with $C_1'(0)$ and $C_1'(-2)$. The target is an additional identity, not a numerical claim that a selected constant basis is complete.

12.3 Colored moving reflection and the Gaussian targets

Replace the kernel by products involving $1 - \xi \mathrm{e}^{-t}$ for roots of unity ξ. Derive the exact reflected colors, branches, and local normalization before extracting harmonic derivatives. A concrete goal is a functional identity whose Gaussian specialization supplies relations outside the already insufficient restricted S6 presentation. Neither a small numerical residual nor the present uncolored centering theorem settles S6 or S8. Their exact frozen coefficient vectors must be retained when testing any new relation.

12.4 Commensurate versus noncommensurate frequencies

For positive rationally related ω_i, determine how much of (10.4) reduces by finite character filters and partial fractions. Compare this with genuinely irrational frequency ratios. The common-zero theorem is uniform in that distinction, but closed evaluations of its derivatives need not be. A first concrete case is two frequencies in ratio $1 : 2$ at multi-depth $(1, 1)$.

12.5 All-order reduction of the shifted logarithmic family

Theorem 5.1 expresses every logarithmic order in Gamma/polygamma, generalized Stieltjes, and Hurwitz spectral coordinates. At rational shifts, determine additional identities among the order derivatives of polylogarithms in (8.8). Such a reduction should be specified in a fixed finite coordinate space and proved; it should not be inferred from an unsuccessful relation search.

12.6 Connection identities for the Borel hypergeometric germ

The function $e^{ct}{}_2F_1(a,b;r+1;1-e^t)$ supplies an explicit analytic germ for every resonant tail. Determine useful connection identities between this germ, the original Gamma ratio, and logarithmic expansions of Gauss's function at its singular endpoint. The objective is exact functional continuation and coefficient transport, not merely estimates for optimal truncation.

12.7 Formalization and integration

The finite proof of Theorem 3.2 is a natural first formal target: it needs Bernoulli finite differences, polynomial divisibility, a degree bound, and one leading coefficient. The tail theorem then adds formal Laurent transforms and uniqueness for a power-series differential equation. For the analytic side, reusable lemmas for normally convergent parameter families, Mellin subtraction, and meromorphic coefficient extraction would support all-depth statements without separate proofs for every tabulated case.

A Algorithms and normalization checks

A.1 Finite construction of every resonant coefficient

For fixed integers $r, k \geq 0$, (4.2) can be evaluated without generating any A_j of index below r: first form $P_r(a)P_r(b)$; then compute the Stirling triangle through row k; finally evaluate the two finite sums in (4.1). The construction uses only rational arithmetic when a, b, c are rational. At a root of either Pochhammer factor, use the polynomial formula directly, not division by the factor. Differentiation at those roots uses (4.11) or direct polynomial differentiation.

For an independent implementation, construct $D_1, \dots, D_{r+k}$ from Bernoulli polynomials and use (3.3). Equality of these two independent constructions is the main exact regression test. The formulas for $Q_{r,1}$ and $Q_{r,2}$ in (4.10) are useful low-order normalization checks.

A.2 Gamma coefficient recursions

If $F(u) = \exp(\sum_{j\geq 1} \eta_j u^j/j) = \sum f_r u^r$, then

$$f_0 = 1, \qquad r f_r = \sum_{j=1}^{r} \eta_j f_{r-j}. \tag{A.1}$$

For the centered Gamma quotient F_c, use

$$\eta_1 = -2L, \qquad \eta_j = \begin{cases} \zeta(j), & j \text{ even}, \\ (2^{1-j}-1)\zeta(j), & j \geq 3 \text{ odd}. \end{cases}$$

For $G(u) = 1/\Gamma(1+u)$, use $\eta_1 = \gamma$ and $\eta_j = (-1)^{j+1}\zeta(j)$ for $j \geq 2$. For $C_r(2)/(\pi^2/2)$, use (7.4). These three recursions must not be conflated: their even zeta coefficients differ.

A.3 Notation crosswalk for repository integration

Our $Z_r(s,a)$ is the family denoted $E_r(s;a-1)$ in [4]. Its heat coefficient $b_j(u,a-1)$ is our $c_j(u,a)$. Our $A_j(a,b,s;c)$ agrees with the Gauss coefficient in [3], and is unrelated to that source's generalized harmonic functions with a similar letter. The centered coefficient $C_r(s)$ is new notation local to this report; it must not be identified with the fixed-shift Gamma polynomial sometimes denoted $C_r(a)$ elsewhere in the manuscript. All theorem and equation labels in this article use the prefix `agrc:`.

B Source and artifact boundaries

The source audit pins the relevant repository content by commit and blob identifiers in the file `audit/source_manifest.json`. It records the exact research question addressed and distinguishes the prior negative-Laurent theorem from the moving-reflection extension. Only inspected textual sources are used as repository evidence; there is no claim to have replayed every incoming ZIP.

The archive contains the standalone article, its shared TeX body, a verification script, exact tables, exact and numerical run records, an integration guide, and a build script. The article can be rebuilt with a standard LaTeX installation; the verification suite uses SymPy and mpmath. Generated log and auxiliary files are not mathematical certificates. A checksum manifest records the delivered artifacts.

References

[1] ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges*, canonical manuscript README and editorial status, inspected at commit `dd78c6c9079e1e4a32600d8f9400667b4de8bd54`, path `Analysis/Polylogarithms/docs/manuscript/README.md`. https://github.com/VladimirReshetnikov/ProveIt.

[2] ProveIt research report, *Stieltjes–harmonic identities*, `sections/further_research.tex`, especially "Negative-balance identities and other summation theorems." Preserved under `Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/stieltjes-harmonic-identities/`, inspected commit as above; blob `1051d5c4348e9b697712468e9c6b0c0f4d6373db`.

[3] ProveIt research report, *Stieltjes–harmonic identities*, `sections/gauss_stieltjes.tex`, in the same report directory, including the successive-subtraction theorem and `gs:residue-rigidity`; blob `d97de05522a0963d7f54b2d0782211f1c5288d39`.

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[5] P. T. Young, *Series of Height One Multiple Zeta Functions*, Integers **24** (2024), A43, 14 pp. DOI: 10.5281/zenodo.11221606. https://math.colgate.edu/~integers/y43/y43.pdf.

[6] P. T. Young, *Global series for height 1 multiple zeta functions*, European Journal of Mathematics **9** (2023), Article 99. Bibliographic comparison as cited in Young's 2024 article.

[7] M. Can, L. Kargın, M. Cenkci, and A. Dil, *On certain harmonic zeta functions*, arXiv:2403.07123 (2024), version 1 inspected. https://arxiv.org/html/2403.07123v1.

[8] NIST Digital Library of Mathematical Functions, Chapter 24, §24.16, *Generalizations of Bernoulli and Euler polynomials*. https://dlmf.nist.gov/24.16.

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